DSML AND DEEP LEARNING · END-SEMESTER PROJECT · NOVEMBER 2026

ApexHunter 2.0

Finding driver mistakes and pit-stop windows in Formula 1 telemetry, and an honest test of how well it works

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Guide: [Guide name]

92	57.9 M	2
sessions	telemetry rows	seasons

THE PROBLEM

Two questions asked of the car's data

1 Where did the driver make a mistake?

Moments where he did something different from the fastest drivers at the same spot, in a way that cost time: a lock-up, an early lift, a missed downshift.

2 When should he change tyres?

From the laps driven so far, a five-lap pit window, and a "pit now" call when a safety car makes the stop cheaper. Checked against the stops drivers really made.

The catch: there is no answer key.
Nobody labels mistakes, and a team's pit lap is a decision, not a correct answer. So every model is scored on tests we can check (planted mistakes, races it never saw) and must beat a simple baseline.

THE PROBLEM

Why this is hard

0.05 s

lost per lap of tyre age, on average

0.9 s

normal spread of lap time from one lap to the next (traffic, wind, small errors)

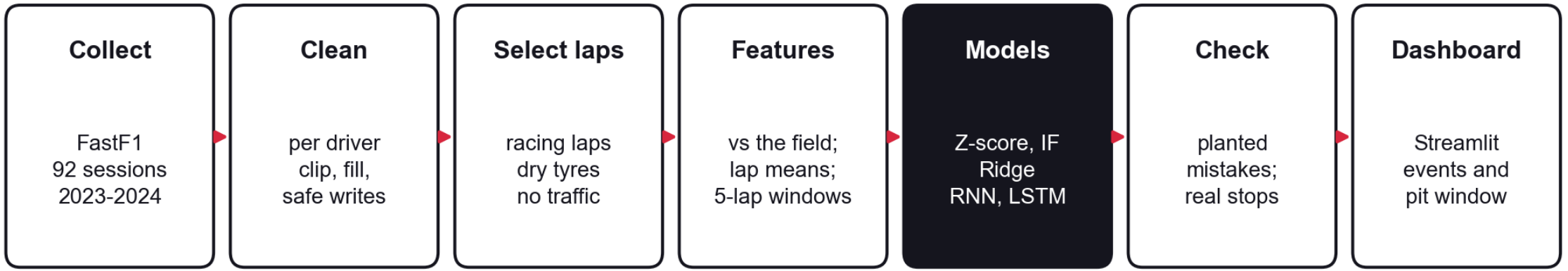
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labelled mistakes in any public F1 data

So the aim is not a perfect strategist.
It is to find what the data really supports, and prove it is not luck, a data leak or overfitting.

WHAT WE BUILT

A pipeline from raw telemetry to a dashboard



Simple baseline

Z-score rule · "no change" (next lap = last lap)

Machine learning (DSML)

Isolation Forest · Ridge regression

Deep learning (DL)

Vanilla RNN · LSTM

DATA

Two seasons of Formula 1 telemetry

92

sessions: every race and qualifying of 2023 and 2024

57.9 M

telemetry rows: speed, throttle, brake, gear, RPM, position

~50,000

race laps with stint, tyre compound, tyre age and pit times

4 / s

samples per second, irregularly spaced (0.014–0.76 s)

Also used: gap to the car ahead and DRS state (to leave traffic out), corner positions (to name where an event happened), track status (safety car, VSC, red flag).

DATA QUALITY

Fixing the data came first

Problem we found	What it would have caused	Fix
Detector compared absolute values	95 % of "mistakes" were ordinary braking	Deviation from the field at the same spot
Pit, safety-car and first laps kept	Slow laps read as mistakes and tyre wear	Racing laps only (one shared filter)
Wet tyres mixed with dry (2,460 laps)	Rain laps read as extreme wear	Dry tyres with known tyre age
Slipstream and dirty air kept	Overtakes flagged as mistakes	Leave out DRS-open or < 1 s behind
Random train/test split of windows	Near-identical windows on both sides	Split by whole races, latest for test
Scaler fitted on all data	Test statistics leaked into training	Scalers fitted on training races only

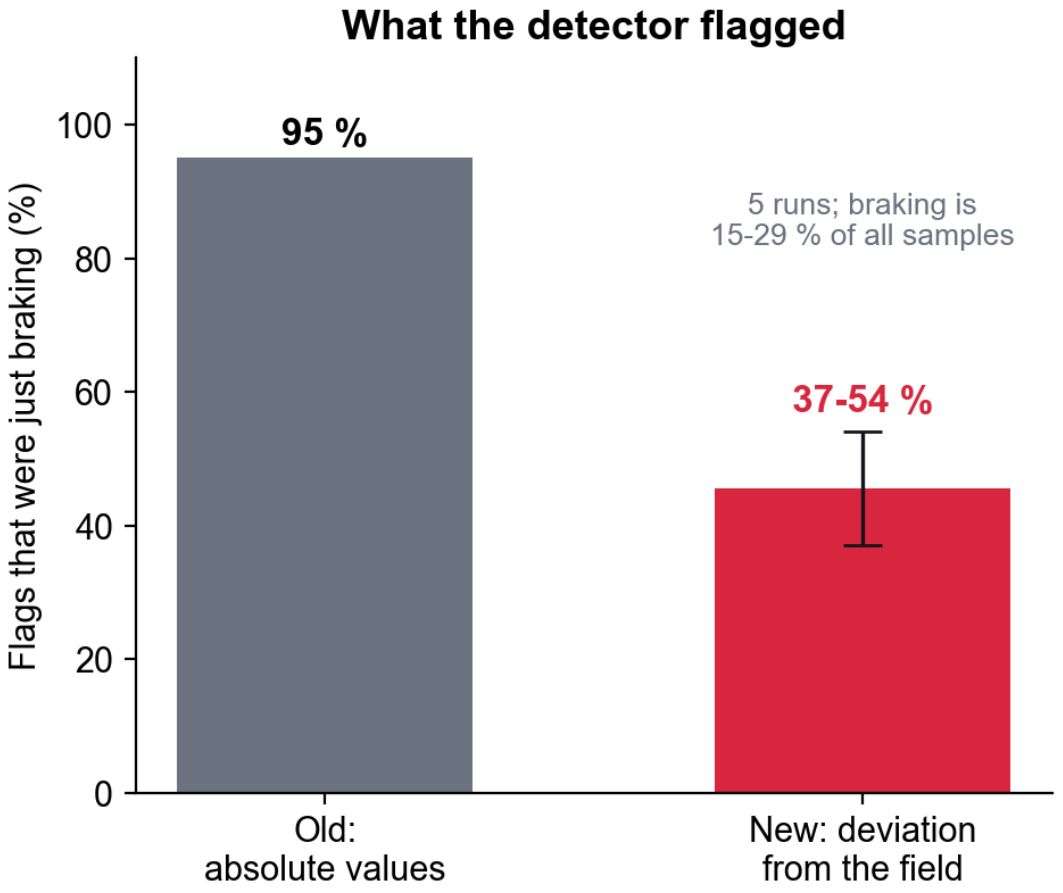
47 problems logged and fixed in total, each with a test; 184 tests, none faking data or models

Compare with the field, not with nothing

Old question: "Is this sample unusual?" Braking is rare, so braking looked like a mistake.

New question: "Is the driver doing something unusual compared with the fastest drivers at this exact point on track?"

Five features, each a deviation from the field per 10 m of track, divided by the field's spread there: speed, throttle, brake, gear, acceleration.



QUESTION 1 · MISTAKES

Two detectors, one test with an answer key

Z-score (baseline)

Flags a sample when any one feature is extreme. No machine learning.

Isolation Forest (DSML)

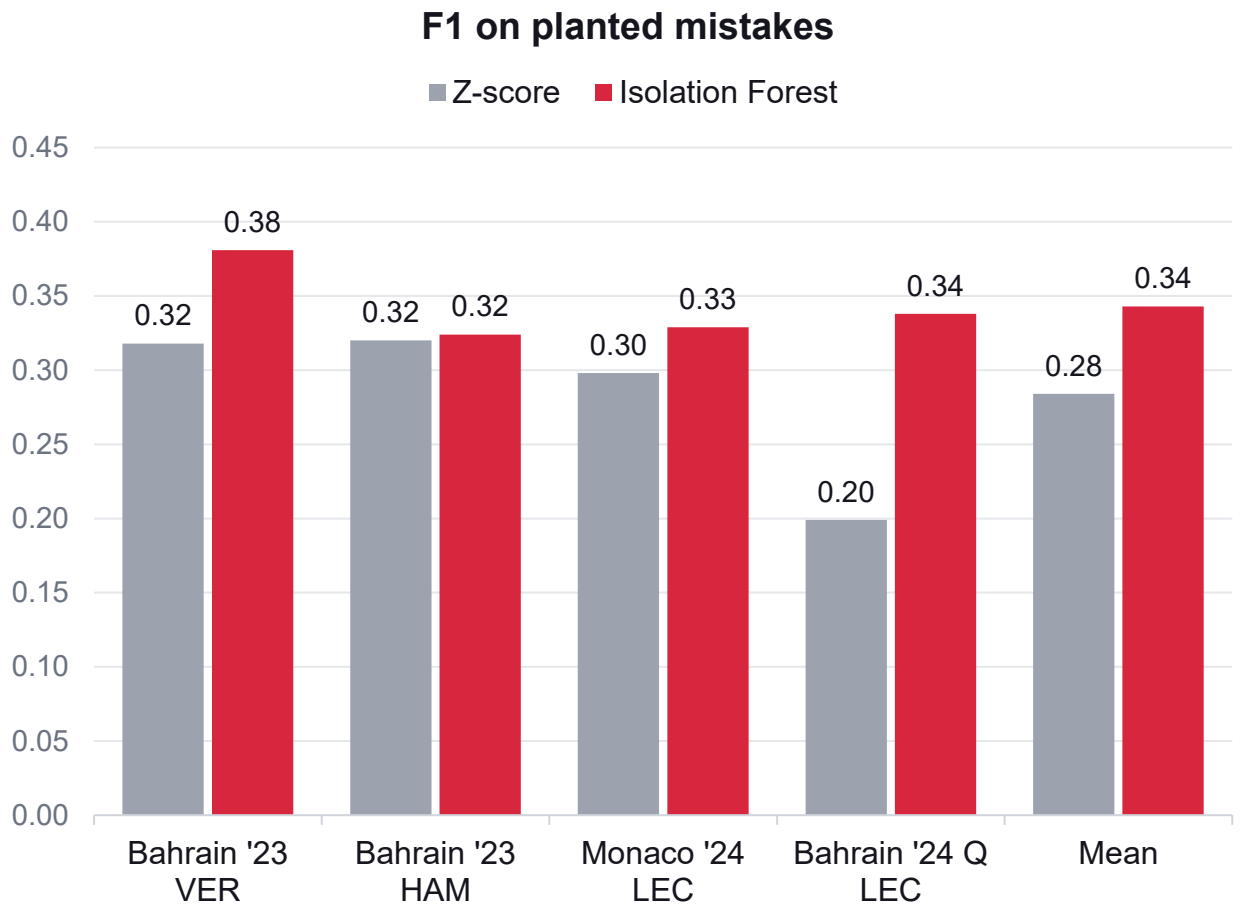
Random trees isolate unusual points in few splits. 200 trees; the 2 % most unusual samples are flagged.

Planted mistake	What is changed
Lock-up	extra 15–30 km/h drop in a braking zone
Early lift	throttle off 0.5–1 s early
Late throttle	throttle held at 0 for 0.5–1 s
Pedal overlap	throttle 20–40 % while braking
Missed downshift	one gear too high in a slow corner

Recall: share of planted mistakes inside an event. **Precision:** share of events on a planted mistake. **Plus a check with no fake data:** do the events cost time against the driver's own laps?

QUESTION 1 · RESULTS

Isolation Forest finds more planted mistakes



0.343

mean F1 (Z-score 0.284)

0.27

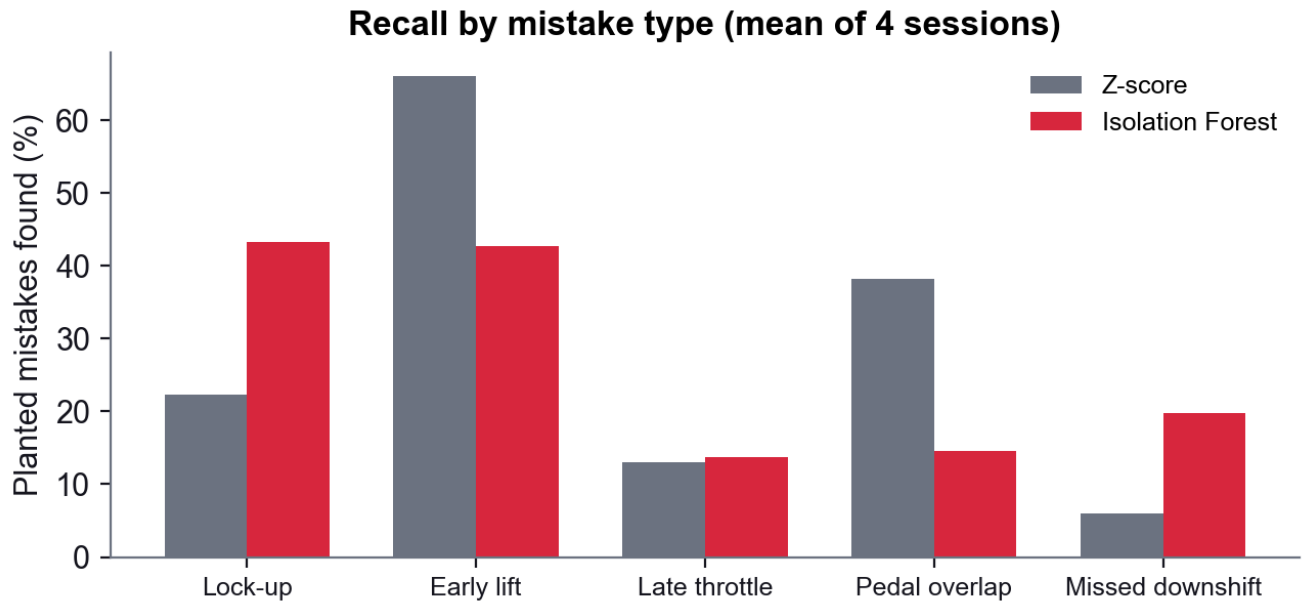
recall: share of planted mistakes found (Z-score 0.20)

15–19 %

of Bahrain events really cost time, vs 10–12 % for random spots

QUESTION 1 · RESULTS

Better where several small things go wrong at once



Top events, Hamilton, Bahrain 2023

1. **Turn 10, lap 38**
   +0.64 s · 16 km/h slower than his own laps
2. **Approaching Turn 12, lap 34**
   +0.52 s
3. **Turn 4, lap 37**
   +0.42 s
4. **Turn 4, lap 47**
   +0.39 s

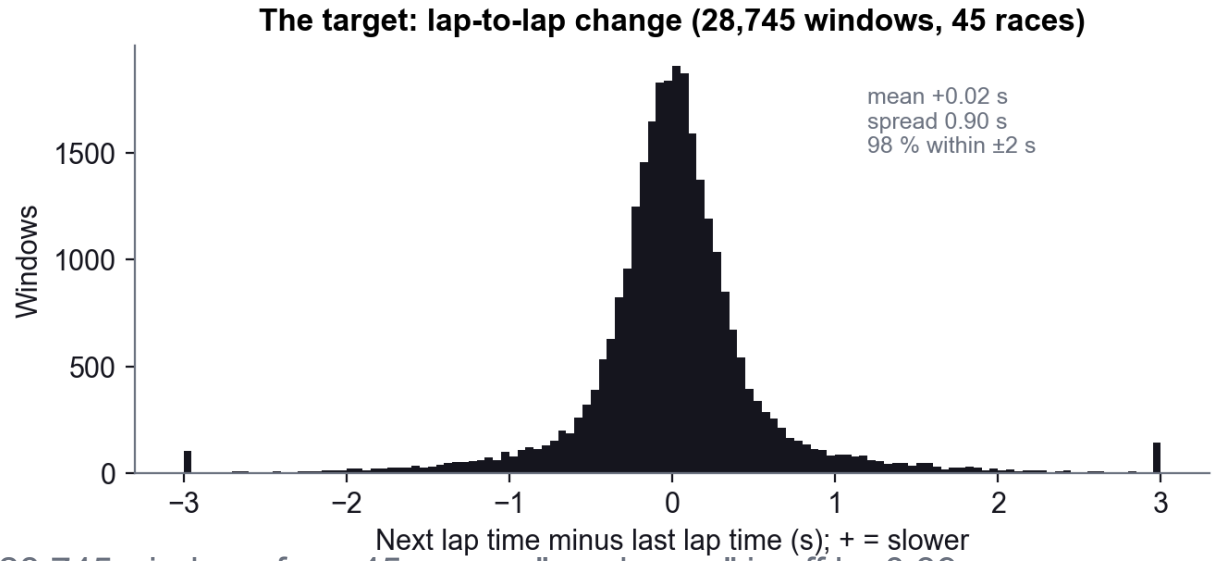
QUESTION 2 · TYRES

Predict how much slower the next lap will be

Inputs: the last 5 racing laps, 9 numbers each: lap time, throttle, brake and RPM (relative to the last lap), tyre age, compound, race lap.

Target: next lap time minus last lap time, in seconds.

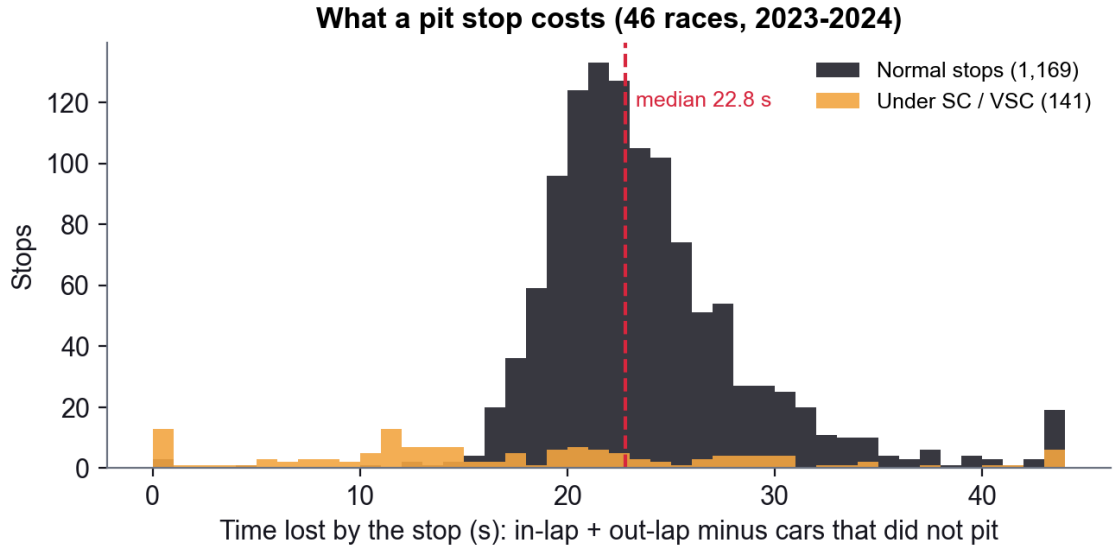
The turning point: predicting absolute speed scored R^2 0.99, but only because every track has its own speed. The change removes the track.



28,745 windows from 45 races · "no change" is off by 0.39 s on average: the bar every model must beat

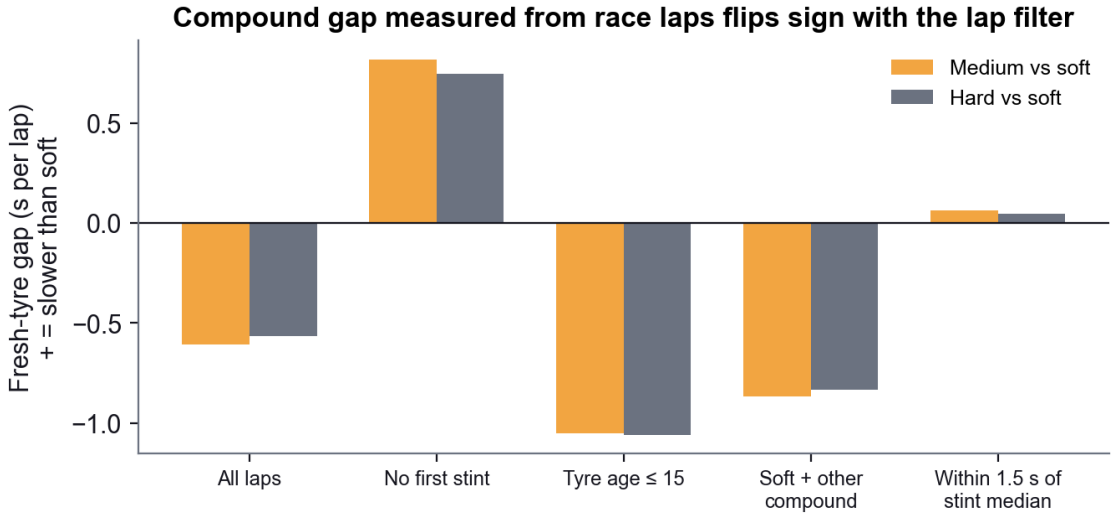
QUESTION 2 · TYRES

What a stop costs, and what the data cannot tell



22.8 s median normal stop (18.5–30 s by track)

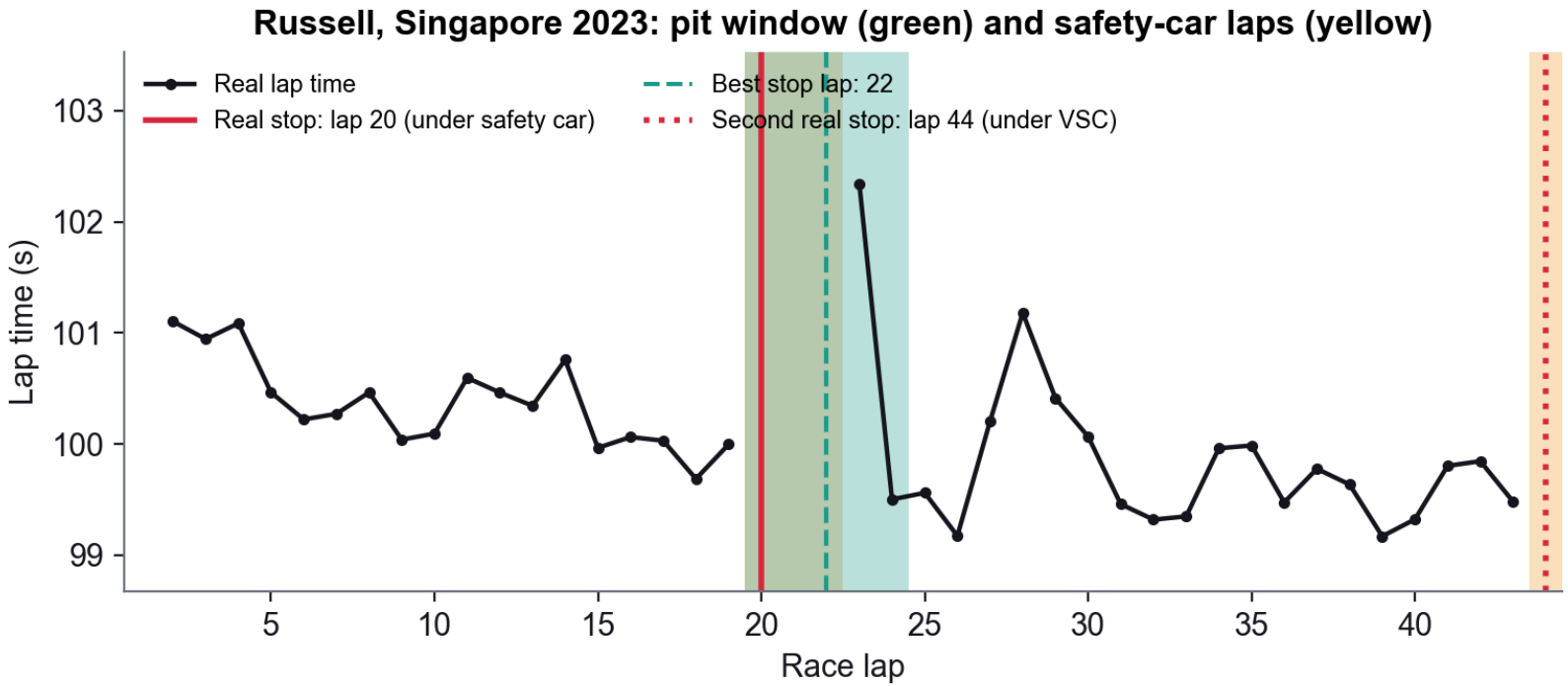
× 0.7 under a safety car or VSC



Compounds look the same in race data: drivers save soft tyres, teams pick the compound for each stint, and "soft" is a different rubber at each track. So no compound advice.

From predicted wear to a five-lap pit window

cost(x) = wear × (old-tyre ages to x + new-tyre ages after x) + pit cost → **window = best x ± 2**



Russell, Singapore 2023

Window laps 20–24, best lap 22. Safety car on lap 20: the stop is 30 % cheaper, so "pit now".

He really pitted on lap 20, inside the window.

DEEP LEARNING

Vanilla RNN vs LSTM

Same windows, same races held out, same metrics and the same pit-window check as Ridge

DEEP LEARNING

Two recurrent networks read the five laps in order

1 Vanilla RNN

$h_t = \tanh(W \cdot x_t + U \cdot h_{t-1} + b)$. One hidden state; gradients can vanish over long sequences.

2 LSTM

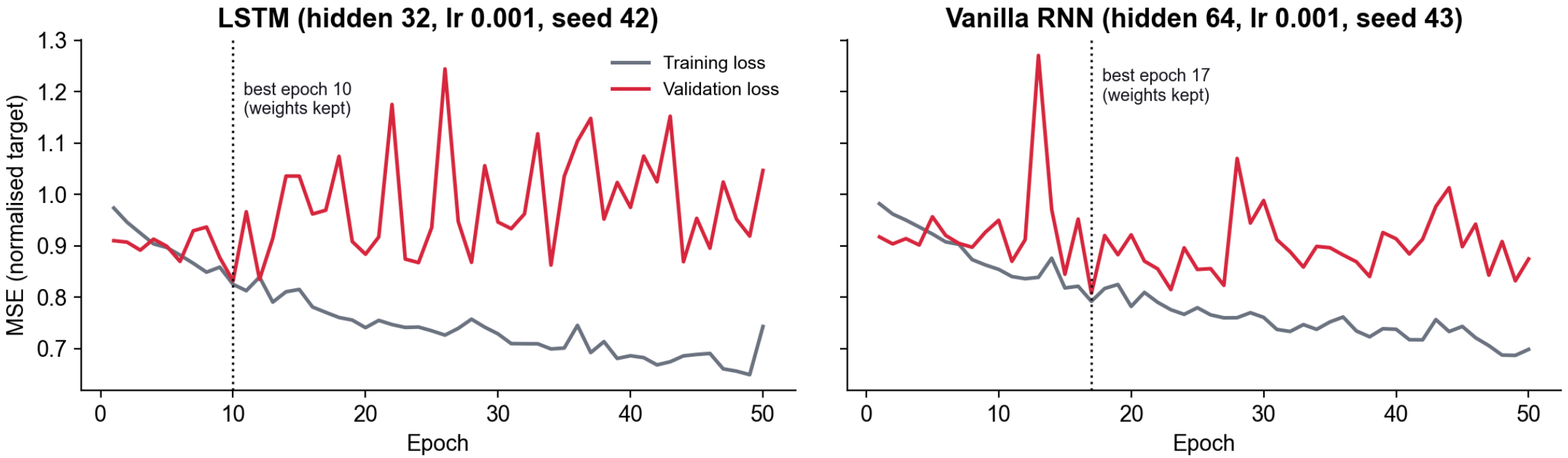
Input, forget and output gates and a cell state carry information across steps without vanishing.

3 Shared setup

2 layers, dropout 0.3, linear head. Adam, MSE loss, batch 32. Grid: hidden {32, 64, 128} × lr.

Training practice: 6-setting search (20 epochs) on the latest 20 % of training races; final model 50 epochs, best-validation weights restored; 3 random seeds, reported as mean ± spread; scalers fitted on training races only.

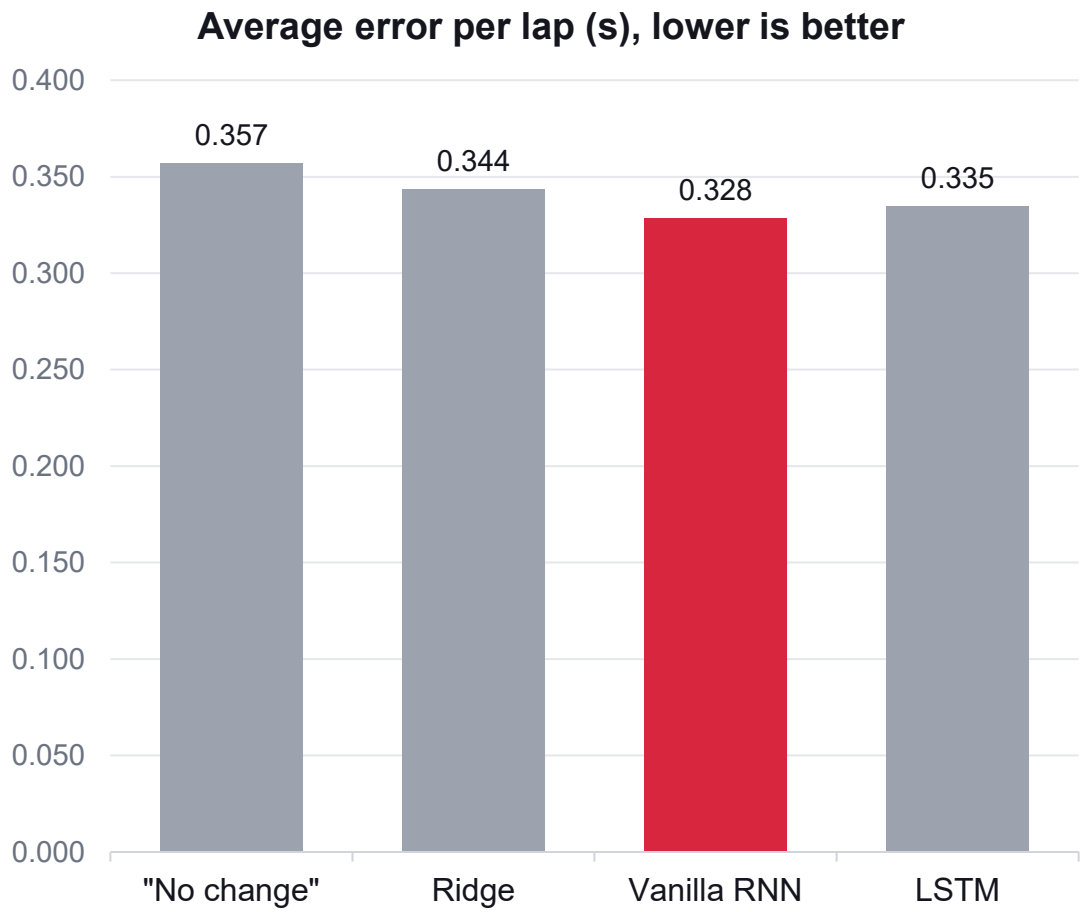
After 10–17 epochs the networks start to memorise

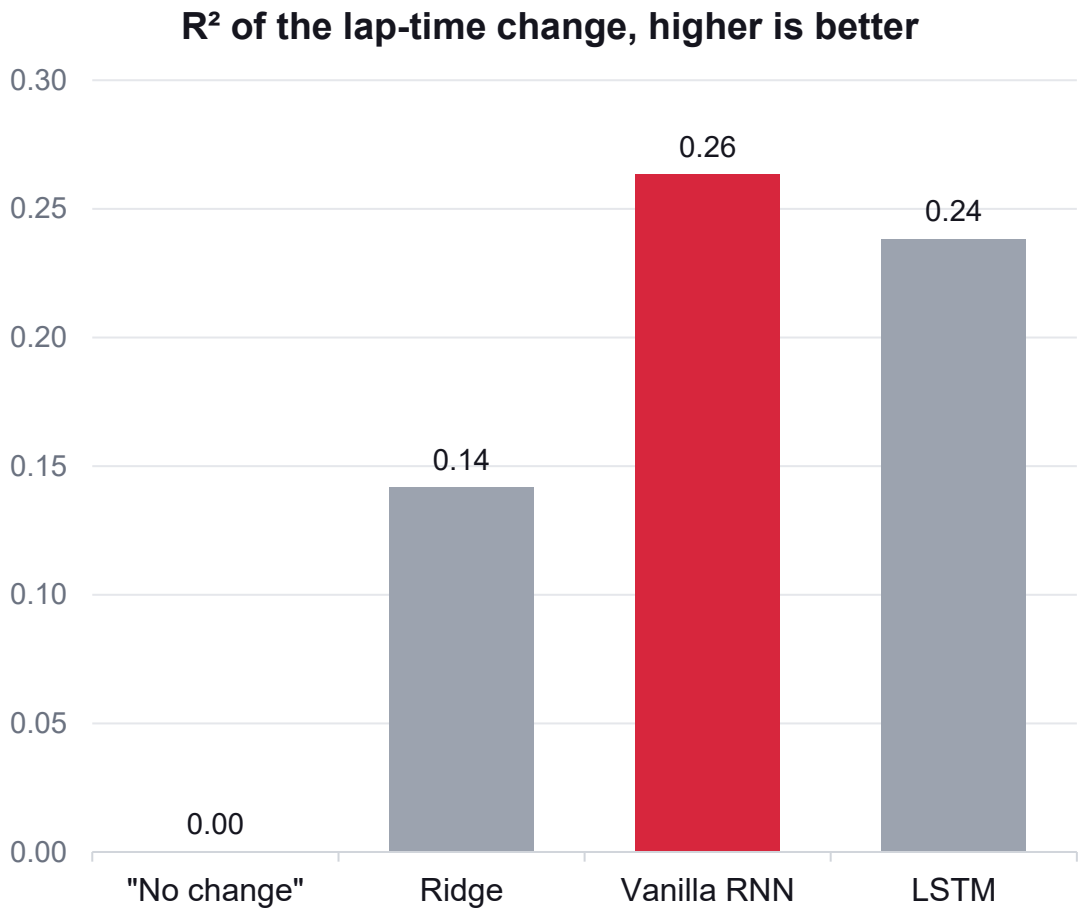


Training error keeps falling; validation error stops improving and becomes noisy. Most lap-to-lap change is noise the inputs cannot explain, so extra training only memorises it. The weights of the best validation epoch are restored.

RESULTS · NEXT-LAP PREDICTION

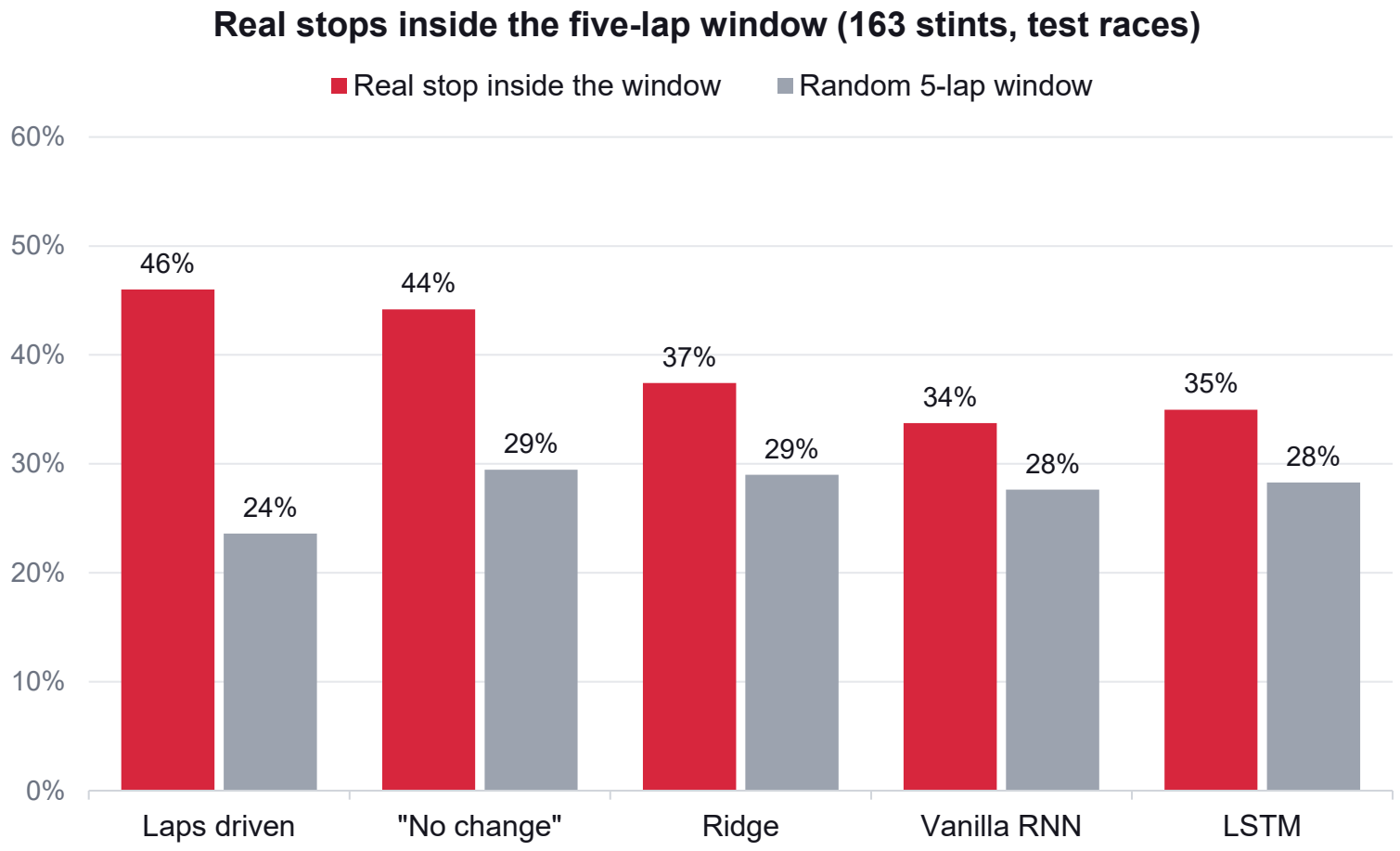
Both networks beat Ridge on lap times





Saved models shown. Mean over 3 seeds: RNN 0.332 ± 0.007 s, LSTM 0.338 ± 0.004 s; with 5-lap sequences the LSTM's gates bring no advantage

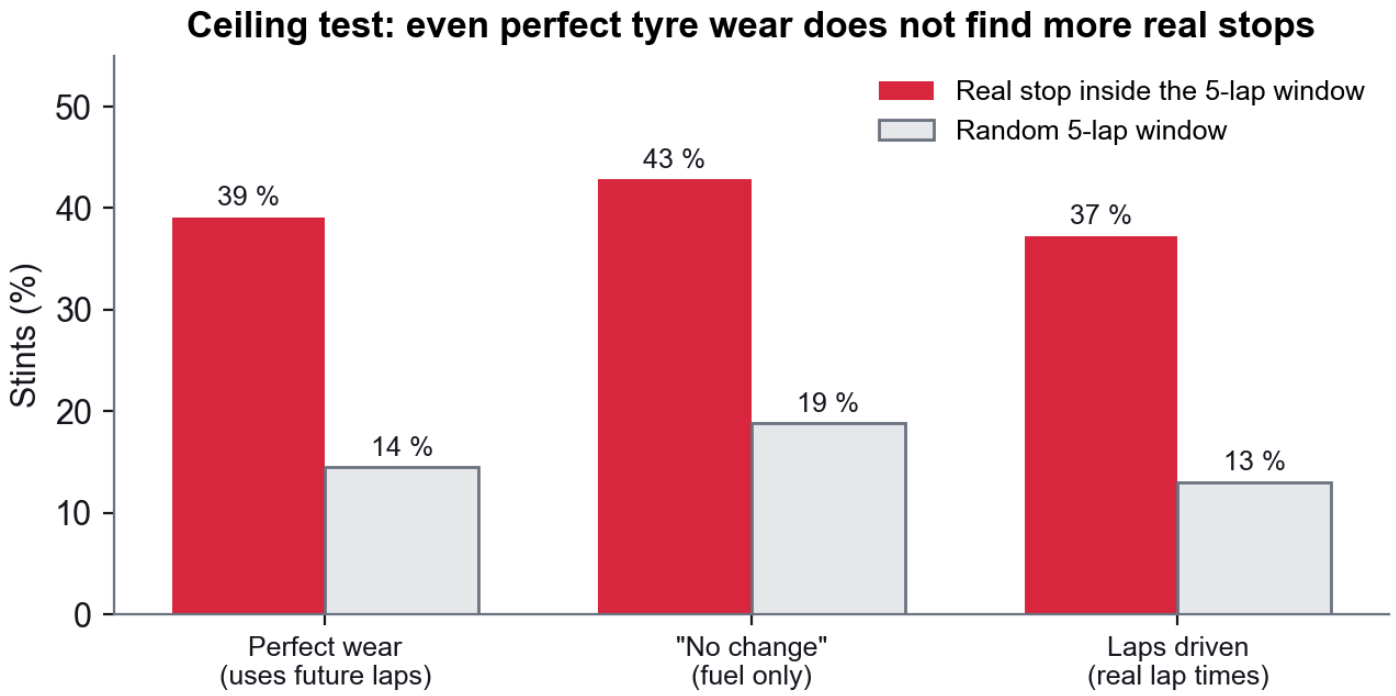
Better lap times do not mean better pit windows



Every source beats chance
about 2× a random window.

The models do not beat the simple sources.
Their predictions correct lap-to-lap noise rather than measure wear; averaged over a stint they often predict laps getting faster, which gives no window.

Even perfect tyre wear would not find more real stops



Why?

Real stops are set by undercuts, traffic, team plans and safety cars.

Wear is so small (0.05 s/lap) that the best stop lap barely moves.

Published work agrees: on F1 tyre data a tree model beat RNN, LSTM, GRU and a transformer (Todd et al., 2025).

THE SYSTEM

Every result is on the dashboard

1 Mistake events

Track map with one marker per event; corner, reason in plain words and time lost; likely lift-and-coast labelled.

2 Pit window

Per stint: lap times, window, safety-car laps, best and real stop; the best wear source and every source's score.

184

automated tests, all real data

47

bugs found and fixed

8

races with pit-window output

1 command

per model: train, check, predict

WHAT WE LEARNED

Six lessons from the project

01 Clean the data first

47 problems would have made later numbers wrong.

02 Compare with the right reference

The field at the same spot turned braking flags into mistakes.

03 Build a test with an answer key

Planted mistakes made an unlabelled problem measurable.

04 Keep a simple baseline

"No change" beat every model on pit windows.

05 Deep learning is not automatic

Networks win on lap times, but the plain RNN ties the LSTM.

06 Report what doesn't work

The ceiling test is the project's most useful result.

LIMITS AND NEXT STEPS

What we can't claim yet

Limitations

Planted mistakes are fake: they show the detector finds these patterns, not that every flag is a real mistake.

Only four sessions scored for mistakes; five test races for tyres.

A tyres-only strategy: traffic, rivals and track position are left out on purpose.

Next steps

LSTM autoencoder for mistakes, scored with the same planted-mistake test (Z-score vs Isolation Forest vs autoencoder).

Predict the laps until a driver pits directly.

Outputs for every driver of every race; dashboard redesign.

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Thank you

Questions are welcome

Mistakes: Isolation Forest F1 0.343 vs Z-score 0.284 · Tyres: RNN and LSTM best on lap times; pit window 2× better than chance